MIL-F-60041B (PA)
14 MARCH 1975
SUPERSEDING
MIL-F-60041A (MU)
6 September 1973

MILITARY SPECIFICATION

FUZE, BASE DETONATING, M578
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of artillery fuze designated as Fuze, Base Detonating, M578.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-116 - Preservation, Methods of

MILITARY

MIL-STD-650 - Explosive: Sampling, Inspection and Testing

DRAWINGS

PICATINNY ARSENAL

FSC: 1390

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8796522 - Marking Diagram and Sealing for Wooden Packing Boxes
8886434 - Fuze, Base Detonating, M578 Loading Assembly
8887101 - Box, Fiber for Fuze, BD, 578
8887102 - Box, Packing, Ammunition for Fuze, BD, M578

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze Assembly. - The fuze assembly shall comply with all requirements specified on Drawing (dwg.) 8886434 and with all requirements specified in applicable specifications.

3.3 Waterproofness. - The fuze assembly shall show no evidence of moisture on the interior surfaces after the test specified in 4.5.1.

3.4 Flatness of Booster Lead Cup. - After loading of the lead charge, booster pellet and cover disc, flatness of the bottom surface of the booster lead cup shall not exceed .002 inches.

3.5 Workmanship. - All parts shall be dry and free of burrs, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - Only quality conformance inspection and testing shall be performed.

4.3 First Article Inspection (None Defined)

4.4 Quality Conformance Inspection.

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4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Fuze parts from one lot interfix number from one manufacturer.

b. RDX for lead charges from no more than two consecutive lots with the same lot interfix number from one manufacturer.

c. RDX for the booster pellets from no more than two consecutive lots with the same lot interfix number from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling Plans. - Unless otherwise specified in the classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9276737
4.4.2.1	Lead Booster Cup -- Prior to Inserting Booster Pellet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major A 101	Lead charge missing, above flush or more than max. below flush		0.40%	3.2	Visual-Gage
Minor 201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886434
4.4.2.2	Lead Booster Cup -- Prior to Assembling to Fuze Body				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major A					
101	Flatness of bottom surface of booster lead cup (after assembling cover disc)		0.40%	3.4	Gage
Major B					
131	Booster pellet missing		0.40%	3.2	Visual
132	Cover disc missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Fuze Assembly, Prior to Assembling Booster Lead Cup Assembly	1	1	8886434
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Shutter in armed position		100%	3.2	Visual
Major B 131	Any component missing		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Fuze Assembly				8886434
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major A					
101	Lead booster cup fails to withstand min. assembly torque		0.40%	3.2	Gage
102	Waterproofness	(a)		3.3	4.5.1
Major B					
131	Surface sealing missing		0.40%	3.2	Visual
132	Alignment screw above flush or stake missing		0.40%	3.2	Visual
133	Thread pilot diameter, max.		0.40%	3.2	Gage
134	Length max.		0.40%	3.2	Gage
135	Thread damaged to extent of preventing use		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:
(a) 4.4.3.1

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Box, Fiber Packing, Prior to Sealing				8887101
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Any packing material missing		0.65%	3.5	Visual
202	Separator damage to extent that fuzes touch each other		0.65%	3.5	Visual
203	Fuze missing		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887101
4.4.2.6	Sealed, Box, Fiber				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major	None Defined				
Minor					
201	Sealing strip torn, bad wrinkled, incomplete or otherwise fails to seal box completely		0.65%	3.5	Visual
202	Carton torn, cut or punctured		0.65%	3.5	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Contents loose		0.65%	3.2	Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887101
4.4.2.7	Sealed Barrier Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major B					
131	Seal improper or incomplete		0.40%	3.5	Visual
132	Bag punctured, torn or cut		0.40%	3.5	Visual
133	Heat seal test		0.40%	3.2	4.5.3
Minor	None Defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packing Box, Prior to Sealing				8887102
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major B 131	Carton missing		0.40%	3.5	Visual
Minor	None Defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887102
4.4.2.9	Packing Box, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major B					
131	Box broken, board loose or protruding nail		0.40%	3.5	Visual
132	Strapping missing, broken or loose		0.40%	3.5	Visual-Manual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.5	Visual
202	Strapping improperly engaged or mislocated		0.65%	3.5	Visual
203	Contents loose		0.65%	3.5	Manual-Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-60041B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Booster Pellet				9276679
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major A 101	Density measurement test	10		3.2	4.5.2
Minor	None Defined				

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4.4.3 Testing

4.4.3.1 Waterproofness (see 3.3) - Major A Defect

4.4.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 50 fuze assemblies (see Dwg. 8886434) shall be selected from each lot for this test. If two or more assemblies fail to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.3.1.2 After three consecutive lots have complied with the criteria of 4.4.3.1.1, 15 sample fuze assemblies shall be selected from each lot for this test. If one or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed herein, is described in the paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Waterproofness. - The fuze assemblies shall be immersed in water to a minimum depth of six (6) inches for a period of not less than four (4) hours. At the end of immersion time, the assemblies shall be removed from the water, wiped dry and dissassembled to determine compliance with the requirements.

4.5.2 Density of Booster Pellet. - The density of the booster pellet shall be determined in accordance with Method 203.1 of Standard MIL-STD-650. If any pellet fails to comply with the requirement, the lot shall be rejected.

4.5.3 Heat Seal of Barrier Bag (see dwg. 8887101). - Test procedures and sampling acceptance criteria for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - The fuze assemblies shall be preserved and packaged in accordance with dwg. 8887101.

5.2 Packing

5.2.1 Level C. - The packaged fuze assemblies shall be packed in accordance with dwg. 8887102.

5.3 Marking. - Marking shall be as specified in dwg. 8887101 and 8796522.

6. NOTES

6.1 Intended Use. - The fuze assembly covered by this specification is intended for use on the M393 complete round.

6.2 Ordering Data. - See MIL-A-48078 (PA).

6.3 Submission of Designs for Approval. - See 6.2 of MIL-A-48078. Submit equipment design, as required, to Commander, Picatinny Arsenal, Attn: SARPA-QA-T-T Dover, New Jersey 07801.

Commander
U.S. Army Picatinny Arsenal
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A-056

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